Math 105L **Lab 5 Part 2**

Fundamental Differentiation Rules & Negative Exponents

The Three Core Differentiation Rules

In this laboratory demonstration, we synthesize the foundational differentiation building blocks:

- **Sum Rule:** The derivative of a sum is the sum of the derivatives:

$$\frac{\mathbf{d}}{\mathbf{dx}}\big(\mathbf{f(x)} + \mathbf{g(x)}\big) = \mathbf{f'(x)} + \mathbf{g'(x)}$$

- **Constant Multiple Rule:** Constants factor cleanly out of differentiation:

$$\frac{\mathbf{d}}{\mathbf{dx}}\big[\mathbf{C} \cdot \mathbf{f(x)}\big] = \mathbf{C} \cdot \mathbf{f'(x)}$$

- **Power Rule:** For any real exponent n:

$$\frac{\mathbf{d}}{\mathbf{dx}}\big[\mathbf{x^n}\big] = \mathbf{nx^{n-1}}$$

Comprehensive Example Worked on Blackboard: $f(x) = \frac{6}{x} + 7x^{10}$

We differentiate $f(x)$ with full pedagogical justification at every algebraic step:

- **Step 1: Rewrite reciprocals as negative powers:**

$$f(x) = \frac{6}{x} + 7x^{10} = \mathbf{6x^{-1}} + \mathbf{7x^{10}}$$

- **Step 2: Apply the Sum Rule:**

$$f'(x) = \frac{d}{dx}\big[6x^{-1}\big] + \frac{d}{dx}\big[7x^{10}\big] \quad \text{(Sum Rule)}$$

- **Step 3: Apply the Constant Multiple Rule (twice):**

$$f'(x) = 6 \cdot \frac{d}{dx}\big[x^{-1}\big] + 7 \cdot \frac{d}{dx}\big[x^{10}\big] \quad \text{(Constant Multiple Rule)}$$

- **Step 4: Apply the Power Rule (twice):**
  - For x^{-1}: $n = -1 \implies \frac{d}{dx}[x^{-1}] = (-1)x^{-1-1} = -x^{-2}$.
  - For x^{10}: $n = 10 \implies \frac{d}{dx}[x^{10}] = 10x^{10-1} = 10x^9$.

$$f'(x) = 6\big(-x^{-2}\big) + 7\big(10x^9\big) \quad \text{(Power Rule)}$$

- **Step 5: Simplify and express with positive exponents:**

$$\mathbf{f'(x)} = -\frac{\mathbf{6}}{\mathbf{x^2}} + \mathbf{70x^9} \qquad \checkmark$$

Key Takeaway for Fractions and Polynomials:
Before applying the Quotient Rule to expressions with constant numerators like $\frac{C}{x^k}$, always rewrite them algebraically as Cx^{-k}. Applying the Power Rule to negative exponents is much faster and significantly less error-prone!